Beyond Task-Specific Entrainment Heads: Mixed Circuits for Lexical-Semantic Relations in LLMs

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Abstract

Large language models are highly capable at in-context learning, but they are also surprisingly easy to mislead with distractor context. Recent work on contextual entrainment argues that this vulnerability is implemented by a set of *entrainment heads* that copy previously seen tokens from context (Niu et al., 2025). Based on single-relation head searches, prior studies have concluded that these entrainment heads are strictly task-specific or relation-specific. In this report, we challenge this interpretation. Using differentiable masking with Gumbel-Sigmoid gates, zero ablation, and cross-domain evaluation on LLaMA-3.1-8B (Louizos et al., 2017; Yu et al., 2024), we demonstrate that characterizing these heads as uniquely relation-specific is inaccurate. Instead, we find that heads discovered for one specific relation transfer robustly to other tasks within the same broad semantic family, aligning with classical structures of lexical semantics such as meronymy and taxonomy (Miller, 1995; Cruse, 1986). Furthermore, our analysis reveals that the recovered head sets are actually *mixed circuits*: alongside these transferable semantic specialists, there exists a subset of domain-general "copy heads" that facilitate context entrainment across entirely unrelated tasks. Therefore, we argue that contextual entrainment is not driven solely by isolated, relation-specific experts, but rather by a combination of a reusable, general context-copying substrate and broader semantic-level specialists.

1 Introduction

Large language models (LLMs) can use contextual demonstrations and prompts to perform tasks without gradient-based updating, a behavior commonly described as in-context learning (Brown et al., 2020). At the same time, this ability creates a robustness problem: models often over-weight nearby contextual evidence, even when it is irrelevant or misleading (Shi et al., 2023; Yoran et al., 2024; Cuconasu et al., 2024; Wu et al., 2024). This issue matters both scientifically, because it bears on how transformers arbitrate between stored knowledge and prompt-local evidence, and practically, because it is closely related to prompt injection, retrieval distraction, and spurious copying from earlier text. Niu et al. (2025) describe one important instance of this behavior as *contextual entrainment*: once a token appears in context, the model becomes more likely to reuse it later, even when it is not the correct answer. Their analysis suggests that the responsible heads are task-specific. We revisit this claim from a different angle. In their setting, "relations" are mostly shallow factual mappings from the LRE benchmark, such as country–capital or country–currency. By contrast, we focus on deeper *Lexical Semantic Relations* (LSRs), including taxonomy and part–whole structure, which require more abstract semantic organization than simple factual slot filling.

Our central hypothesis is that LLMs organize relational knowledge not only as isolated factual mappings, but also at the level of broader semantic families. Under this view, task-conditioned head sets should be mixed circuits: some heads capture family-level structural regularities, while others form a weaker but more distributed entrainment-like component that transfers more broadly across domains.

2 Related Work

Our work sits at the intersection of distraction in prompting and mechanistic interpretability. On the prompting side, prior research has shown that LLMs can be diverted by irrelevant or misleading context, especially when that context is formatted like a plausible answer (Shi et al., 2023; Yoran et al., 2024; Cuconasu et al., 2024; Wu et al., 2024). In the retrieval setting, this problem is often framed as robustness to noisy or off-topic evidence. That framing is useful, but it does not by itself explain *how* a model becomes distracted.

Mechanistic interpretability offers tools for answering that question. Transformer-circuit work has shown that attention heads can implement surprisingly crisp computational roles, including induction-like copying behaviors and more specialized circuits for specific tasks (Elhage et al., 2021; Olsson et al., 2022; Wang et al., 2023). More recent work has applied differentiable circuit-discovery methods to real language-model computations (Yu et al., 2024). The closest work to ours is Niu et al. (2025), who identify contextual entrainment and propose entrainment heads as its mechanism. We build directly on their framing, but our contribution is to separate *relation-specific semantic heads* from a more distributed *cross-domain entrainment circuit*. In this sense, our report refines rather than rejects the original mechanistic story.

3 Experimental Setup

3.1 Model and intervention pipeline

We used LLAMA-3.1-8B as the main model and implemented all interventions in TransformerLens-style attention hooks. We also piloted the pipeline on LLAMA-3.2-1B, but all quantitative results reported below come from the 8B model. The model has 32 layers with 32 heads per layer, for a total of 1024 attention heads.

To search for important heads, we followed a differentiable masking procedure inspired by recent circuit-discovery work (Yu et al., 2024) (Niu et al., 2025) . For each attention head (ℓ, h), we assigned a learnable gate $z_{\ell,h} \in \{0, 1\}$ applied to the head output at `attn.hook_z`. During optimization, the binary gates were relaxed with a Gumbel-Sigmoid straight-through estimator:

$$\tilde{z}_{\ell,h} = \sigma\left(\frac{\alpha_{\ell,h} + g_{\ell,h}}{\tau}\right),$$

where $\alpha_{\ell,h}$ is the learnable mask logit, $g_{\ell,h}$ is Gumbel noise, and τ is the temperature. The search objective combined an effect term that favored the gold token over the distractor token with a sparsity term encouraging fewer active heads.

After training the mask, we exported the discovered circuit as the $k = 20$ heads with the lowest learned mask values. We then performed *zero ablation* by setting those heads' z outputs to zero at inference time.

3.2 Prompt construction and evaluation

Our prompts followed the attack pattern used and closely related to the contextual-entrainment setup in Niu et al. (2025).We specifically employ factual distractors rather than counterfactuals to isolate the pure effect of contextual entrainment; by presenting a true statement about a disjoint entity, we ensure that any erroneous copying is driven solely by structural mimicry rather than a forced conflict with the model's internal parametric knowledge. An attacked prompt prepended a structurally parallel answer before the query. For example:

```
The capital of Japan is Tokyo.  The
capital of China is
```

Here, `Tokyo` is the distractor, while `Beijing` is the gold answer. The corresponding clean prompt removes the attack sentence:

```
The capital of China is
```

We measured *logit margin*

$$m(x) = \ell(y_{\text{gold}} \mid x) - \ell(y_{\text{dist}} \mid x),$$

and the quantity of interest was the ablation-induced change

$$\Delta m(x) = m_{\text{ablated}}(x) - m_{\text{base}}(x).$$

Under attacked prompts, positive Δm means that ablating the selected heads makes the model less willing to copy the distractor and more willing to recover the correct answer.

We evaluated four relation families in the main cross-domain study: country–capital, object–color, profession–workplace, and sport–equipment. We also ran two structured generalization studies designed to probe semantic organization more closely. These structured splits were chosen to test our hypothesis that relational knowledge in LLMs may be organized at the level of broader lexical-semantic families, rather than only as isolated factual mappings:

1. a **taxonomy** split from natural categories to artificial categories; and

2. a **fine-grained part–whole** split from component–integral to member–collection and portion–mass.

All relations were filtered to keep single-token answers, matching the evaluation code in the notebooks and simplifying the interpretation of token-level logit margins.

4 Results

4.1 Single-relation search returns mixed circuits, not universal heads

We first ran differentiable masking on country–capital attacks. The resulting 20-head circuit represented only 1.95% of all heads, but ablating it still raised the attacked margin by $+0.552$ on the same task. This confirms that the discovery method is finding a causally meaningful subnetwork rather than diffuse noise.

More importantly, this circuit also showed nontrivial transfer beyond the relation on which it was discovered. When we applied the same 20-head set to other relation families, it still improved the attacked margin by $+0.101$ on object–color and by $+0.872$ on profession–workplace, although it hurt performance on sport–equipment (-0.166). These results suggest that heads discovered from one relation can indeed remain effective on other relations, providing evidence of at least partial generalization. At the same time, the transfer is clearly uneven rather than uniform. Therefore, the discovered circuit should not be interpreted as a universal fact-retrieval module; instead, it appears to contain a mixture of components, some of which generalize across relations while others remain task- or family-dependent.

4.2 Relation-specific experts emerge at the semantic-family level

These structured generalization experiments provide direct support for our main hypothesis: LLMs do not organize relational knowledge only as isolated factual mappings, but also at the level of broader semantic families. In taxonomy, the 20 heads discovered on natural-kind classification transferred strongly to artificial-kind classification, with attacked Δ margin increasing from $+1.275$ in-domain to $+2.909$ cross-domain. This suggests that the discovered circuit is not tied to one specific content domain, but instead captures a more abstract taxonomic computation.

The part–whole experiment reveals an even more informative pattern. Heads discovered on component–integral relations generalized strongly to member–collection ($+2.146$) but only weakly to portion–mass ($+0.352$). This implies that the model does not treat all part–whole relations as equivalent. Rather, it groups together some relations that share deeper structural regularities while separating others that are linguistically and conceptually distinct. In this sense, semantic computation appears to be organized at the level of relation families rather than individual factual pairings.

This interpretation is reinforced by the strongest single-head results. Among the 20 component-discovered heads, `L5H17` had a large effect on component–integral ($+0.345$) and member–collection ($+0.456$), but only a modest effect on portion–mass ($+0.112$). Similar patterns appeared for other high-impact heads such as `L15H6`. These heads are therefore best understood as *relation-specific experts*: not general-purpose copy heads, but components of circuits tuned to particular lexical-semantic computations. Taken together, these results support our broader claim that task-conditioned head discovery can reveal circuits specialized at the semantic-family level, rather than only at the level of shallow surface relations.

4.3 Weak cross-family transfer suggests a mixed circuit beyond semantic-family specialists

If the heads discovered by task-conditioned masking consisted only of semantic-family specialists, then no individual head should transfer much beyond its home family. Our single-head analysis suggests a more mixed picture. Head `L9H6`, although much weaker than `L5H17` on component–integral, remained positive across all three fine-grained part–whole variants: $+0.065$ on component–integral, $+0.070$ on member–collection, and $+0.158$ on portion–mass. Unlike a specialist head, its effect was not concentrated on one relation subtype.

We then directly compared `L9H6` against `L5H17` over five unrelated task families. Table 2 shows the result. `L5H17` behaved like a semantic-family specialist: it remained strongest on component–integral and was near-zero or negative elsewhere. By contrast, `L9H6` was much weaker, but also somewhat broader, with small positive effects on profession–workplace and component–integral while remaining close to zero on the other tasks. These results do not establish a universal head. Instead, they suggest that once family-specific experts are separated out, a weaker cross-family component remains. This is consistent with our hypothesis that discovered head sets are mixed circuits, containing both semantic-family specialists and more distributed entrainment-like components.

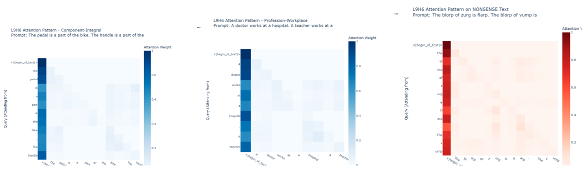
Qualitative attention patterns and a preliminary activation-patching probe were broadly consistent with a distributed entrainment-like component

Discovery source	Evaluation relation	Attacked Δ margin
Country–Capital	Country–Capital	+0.552
Country–Capital	Object–Color	+0.101
Country–Capital	Profession–Workplace	+0.872
Country–Capital	Sport–Equipment	−0.166
Taxonomy-A (natural kinds)	Taxonomy-A (in-domain)	+1.275
Taxonomy-A (natural kinds)	Taxonomy-B (artificial kinds)	+2.909
Component–Integral	Component–Integral (in-domain)	+3.911
Component–Integral	Member–Collection (cross-domain)	+2.146
Component–Integral	Portion–Mass (cross-domain)	+0.352

Table 1: Family-level generalization results from the uploaded notebooks. The pattern is neither fully task-specific nor fully universal. Instead, heads discovered on one relation often transfer strongly *within* a structural family, while transfer across unrelated families is much weaker or inconsistent.

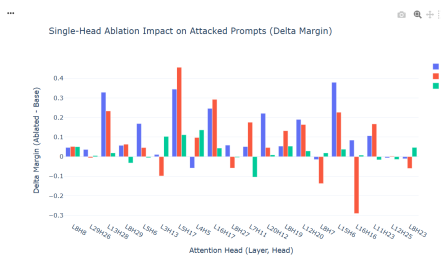
Relation	Single head `L9H6`	Single head `L5H17`	Top-5 multi-head entrainment circuit
Country–Capital	−0.004	−0.010	+0.180
Object–Color	−0.017	−0.041	+0.474
Profession–Workplace	+0.119	+0.034	+2.765
Sport–Equipment	−0.039	+0.032	+0.442
Component–Integral	+0.065	+0.345	+1.523

Table 2: Single-head duel versus multi-head ablation. `L5H17` behaves like a semantic-family specialist, whereas `L9H6` provides only weak evidence of broader transfer. The strongest support for a reusable entrainment-like mechanism comes from the distributed top-5 circuit, not from any single head alone.



(a) Semantic Tasks (Part 1) (b) Semantic Tasks (Part 2) (c) Gibberish Control

Figure 1: Qualitative evidence for a cross-domain entrainment-like head. In the notebooks, `L9H6` responds to similar prompt structure across distinct relations, including nonsense text.



We also attempted a causal activation-patching probe by injecting the final-token activation of `L9H6` from a country–capital prompt into an object–color prompt. In the current experiment, this single-site patch did *not* fully hijack the top-1 prediction. That negative result is informative: it suggests that the broader component is distributed and redundant, rather than concentrated in a single activation vector. This directly motivates the multi-head analysis below.

4.4 Multi-head ablation reveals the distributed component predicted by our hypothesis

The strongest support for our hypothesis comes from the multi-head analysis. Our claim is not that semantic-family specialists disappear; Section 4.2 shows that they clearly exist. Rather, our hypothesis predicts that task-conditioned head sets should also contain a more distributed component that is not confined to a single semantic family. To test this, we scored each candidate head by its average attacked Δ margin across all five task families. The top five heads under this criterion were `L15H10`, `L12H12`, `L16H3`, `L16H17`, and `L16H30`.

Ablating this five-head circuit produced a qualitatively different outcome from ablating any one head alone. The attacked Δ margin rose to +0.180 on country–capital, +0.474 on object–color, +2.765 on profession–workplace, +0.442 on sport–equipment, and +1.523 on component–integral. Unlike the semantic-family specialists discussed earlier, this circuit disrupted distractor-following across semantically unrelated task families. We therefore interpret it not as a replacement for semantic-family specialization, but as the distributed entrainment-like component that coexists with it.

Taken together, Sections 4.2–4.4 support our main hypothesis. We additionally verified that the main qualitative patterns were stable across prompt templates and random seeds, suggesting that the reported effects are not artifacts of a single template choice or initialization.

5 Discussion

Our results support the main hypothesis of this report: relational computation in LLMs is not organized only as isolated factual mappings, but partly at the level of broader semantic families. This is most visible in the structured transfer results. Heads discovered on taxonomy transfer across taxonomic domains, and heads discovered on component–integral transfer much more strongly to member–collection than to portion–mass. These patterns suggest that some heads are best understood as *relation-family specialists*, capturing structural regularities shared within a lexical-semantic family rather than memorizing one shallow factual relation.

At the same time, the discovered head sets are not homogeneous. Beyond these semantic-family specialists, our experiments also point to a weaker but reusable entrainment-like component. The single-head analysis of `L9H6` provides only suggestive evidence for this broader component, but the multi-head ablation results make it much clearer: a jointly selected five-head circuit disrupts distractor-following across five semantically unrelated task families. This implies that task-conditioned masking does not return a single kind of object. Instead, it recovers *mixed circuits* containing both family-specific semantic heads and a more distributed prompt-following substrate.

This interpretation also helps explain why `L9H6` was interpretable but weak. In our notebooks, it functioned as a useful probe because its behavior was visible across domains, but it was not by itself a decisive control point. The stronger evidence came only at the circuit level. This is consistent with redundancy and self-repair effects reported in prior mechanistic interpretability work (McGrath et al., 2023).

The main limitations of this study are straightforward. First, the prompts are synthetic and constrained to single-token answers. Second, the qualitative evidence from attention patterns is suggestive rather than definitive. Third, the activation patching result was preliminary and did not achieve a full top-1 output hijack. Future work should therefore test the same hypothesis on larger factual datasets such as LRE (Hernandez et al., 2024), with stronger causal interventions and more natural prompting formats.

6 Conclusion

This report revisited the claim that entrainment heads are task-specific units. Our results suggest a more precise interpretation. The model does contain heads that behave like semantic-family specialists, especially in taxonomy and fine-grained part–whole computation. But task-conditioned head discovery also reveals a second component: a distributed entrainment-like circuit that supports prompt-conditioned copying across unrelated tasks.

The broader conclusion is therefore not simply that entrainment heads are task-specific or universal. Rather, relational computation in LLMs appears to be mechanistically decoupled: part of it is organized at the level of lexical-semantic families, while part of it relies on a reusable cross-domain prompt-following substrate. This mixed-circuit account fits our transfer, single-head, and multi-head results better than a purely task-specific interpretation, and provides a more useful starting point for future interpretability and robustness interventions.

7 Statement of Contributions

Jinghan Zheng led the overall project conceptualisation, formulated the core hypothesis of mixed circuits versus strictly relation-specific heads, developed the entire codebase for differentiable masking with Gumbel-Sigmoid gates and zero ablation on LLaMA-3.1-8B, executed the primary cross-domain evaluations, and drafted the core analytical sections of the paper (Abstract, Methodology, and Results).

Yuzhe Wang supported the experimental pipeline by constructing the factual attack prompt datasets, generating the attention heatmaps and empirical plots from the raw evaluation logs, and drafting the prompt construction subsection.

Yunjia Tian run the random pruning experiment for stronger enidence. contributed by conducting the literature review on lexical semantics, assisting with the qualitative analysis of the identified attention heads, drafting the Introduction, and formatting the bibliography.

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